



Hepatic and biliary Fascioliasis

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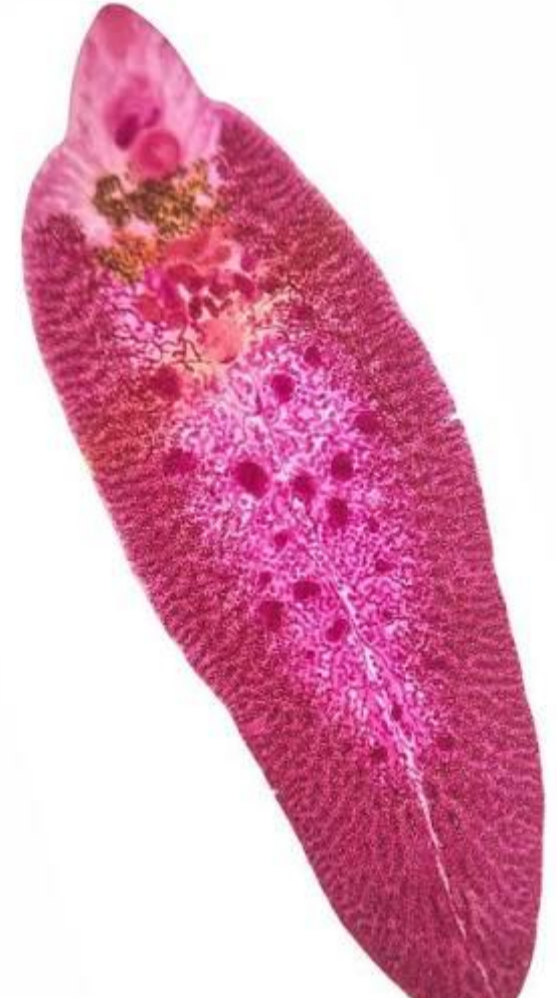
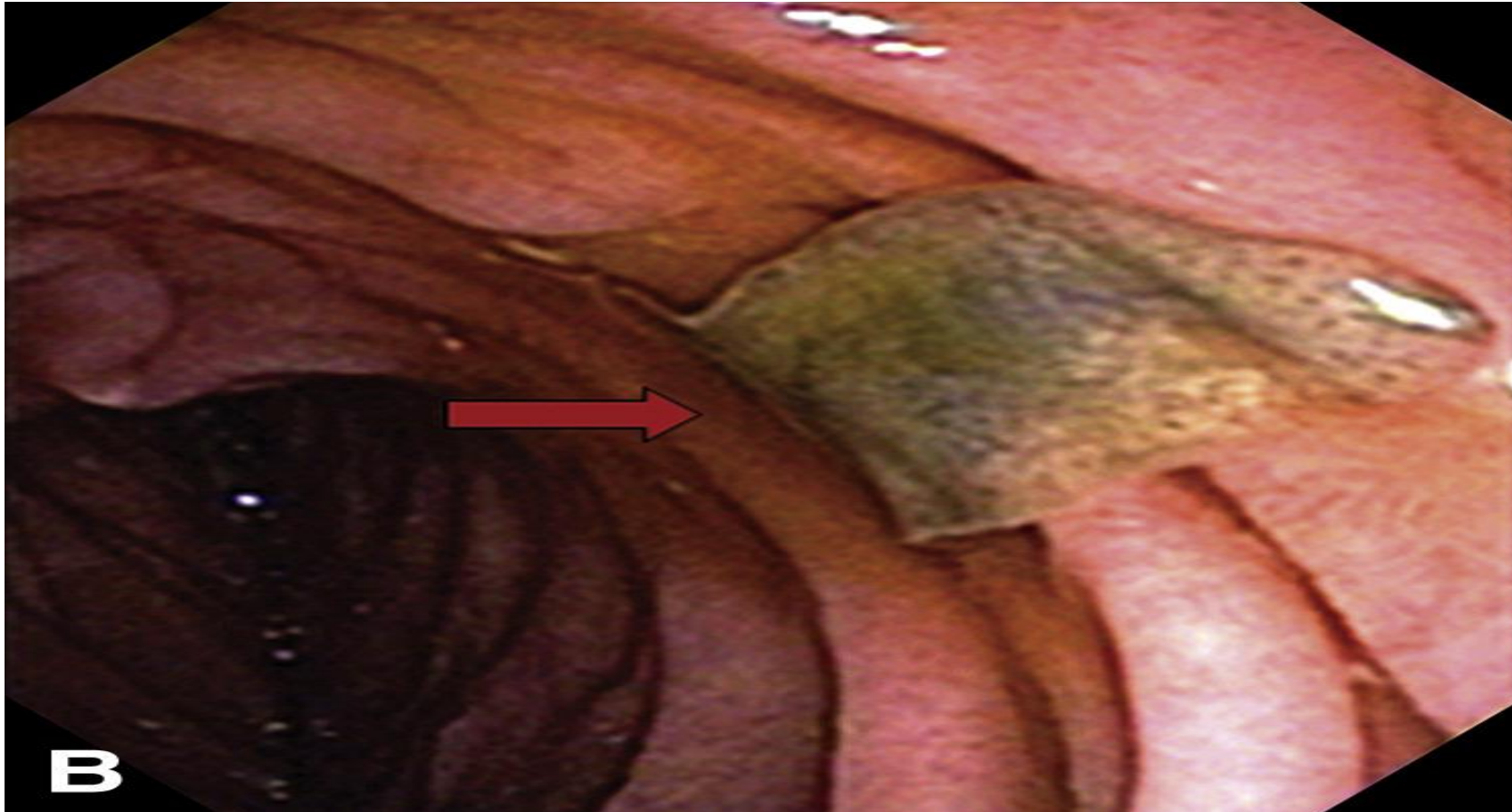
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Third Stage

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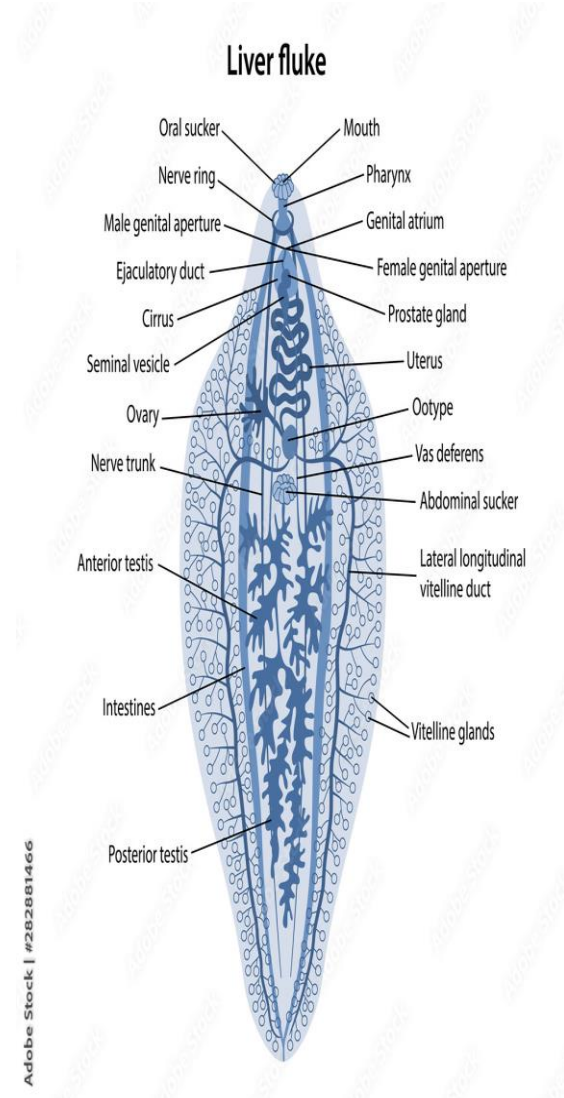
Fasciola Hepatica

- Common name (Sheep liver fluke)



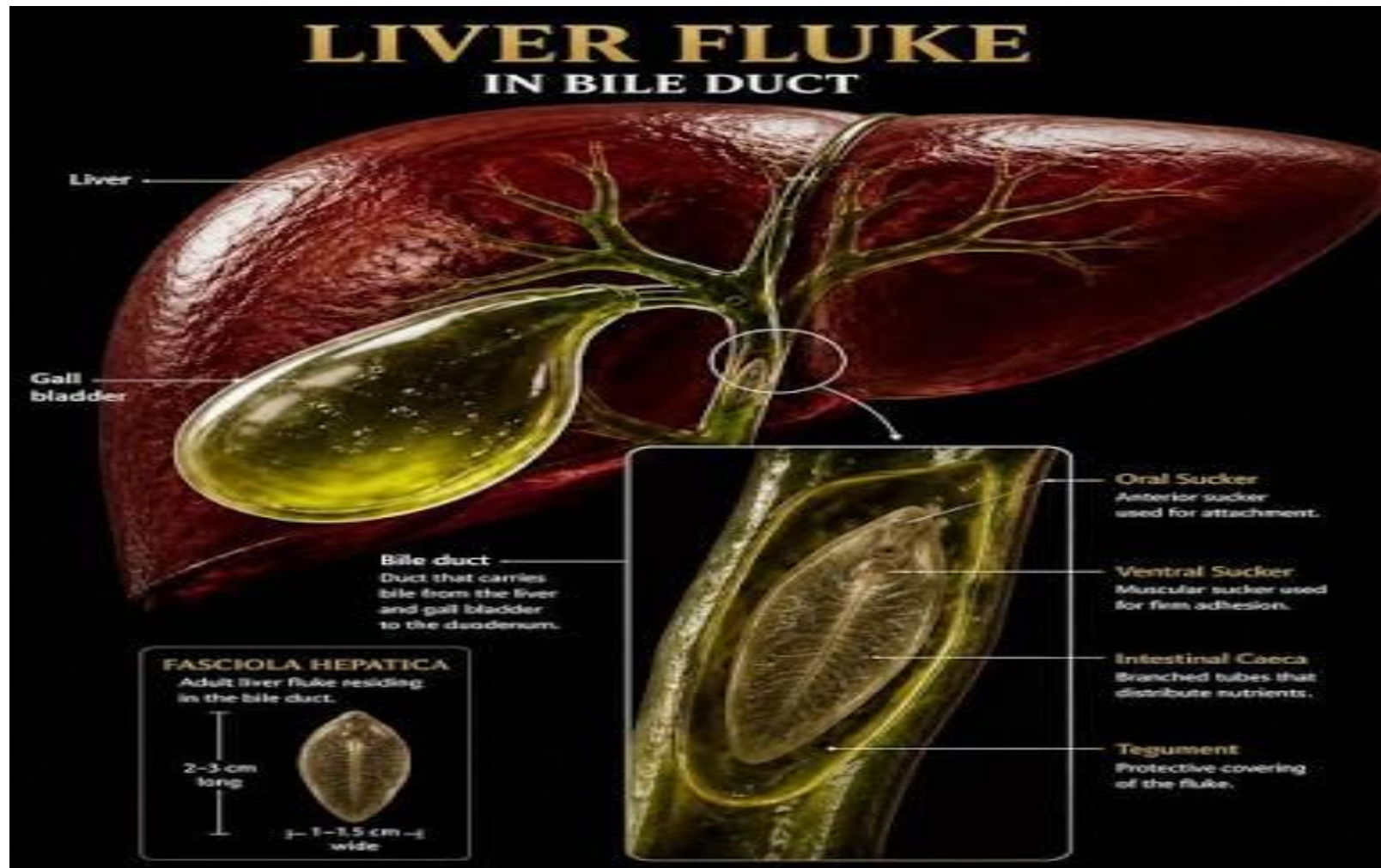
Distribution

- It is the largest and most common liver fluke found in humans, but its primary host is the sheep and to a less extent, cattle.
- It is worldwide in distribution, being found mainly in sheep-rearing areas.
- It causes the economically-important disease, ‘**liver rot**’, in sheep.



Habitat

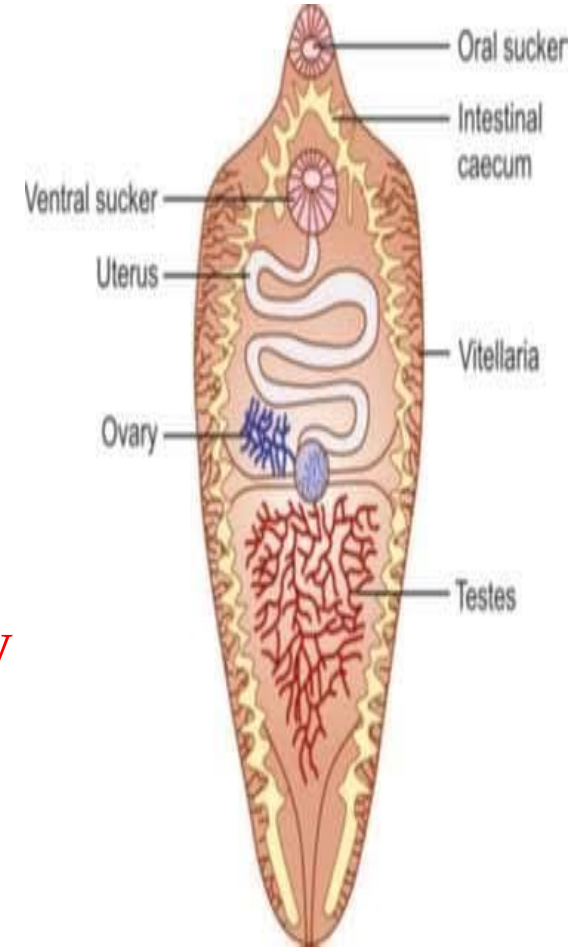
The parasite resides in the **liver and biliary passages** of the definitive host.



Morphology

- **Adult Worm**

- It is a large leaf-shaped fleshy fluke, 30 mm long and 15 mm broad, grey or brown in color.
- It has a conical projection anteriorly containing an oral sucker and is rounded posteriorly
- The adult worm lives in the biliary tract of the definitive host for many years—about 5 years in sheep and 10 years in humans.
- it is hermaphrodite.



Morphology

Egg: The eggs are large, ovoid, **operculated, bile-stained(yellow-brown)**, and about 140 μm by 80 μm in size.



Life Cycle:

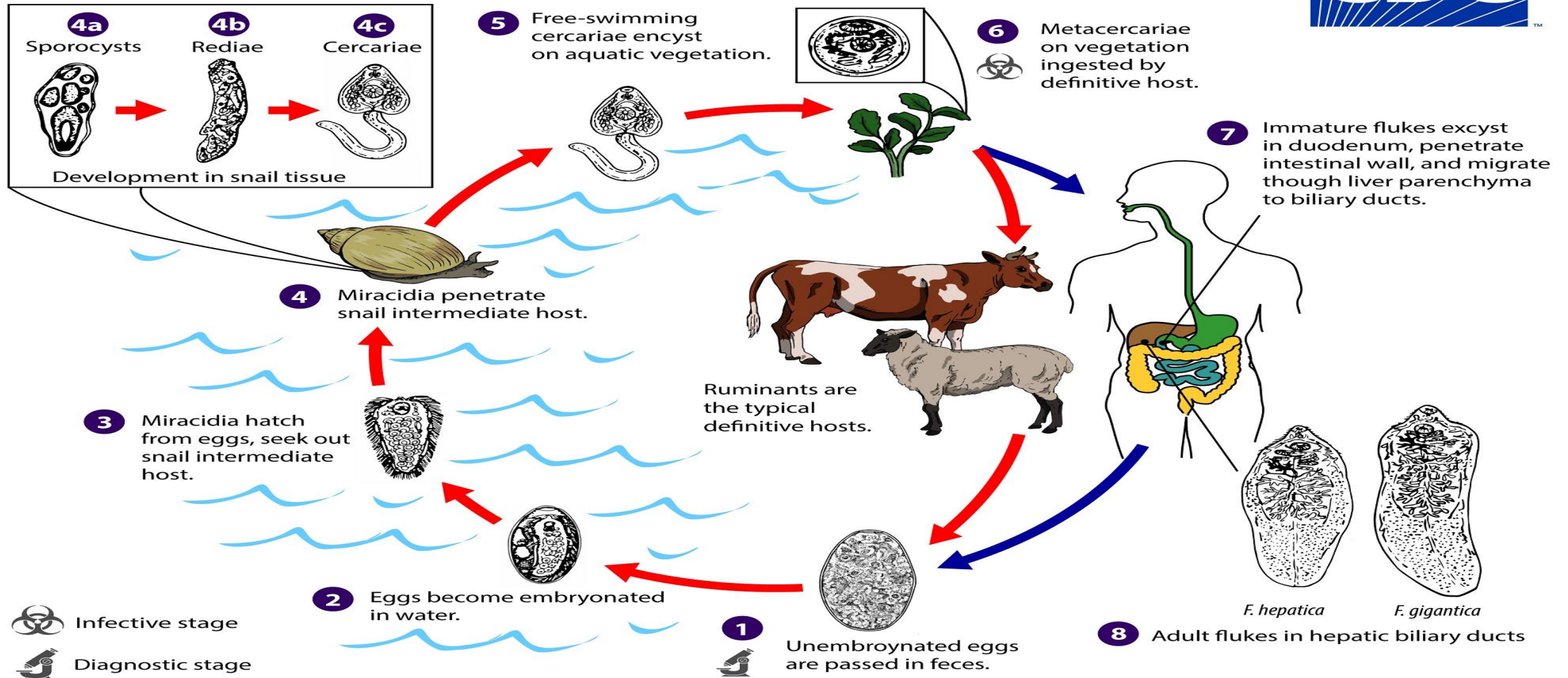


F. hepatica passes its life cycle in:

- 1 definitive host and 2 intermediate hosts.
- Definitive host: Sheep, goat, cattle, and human.
- First intermediate host: Snails.
- Second intermediate host: aquatic plants
- Infective stage: metacerceriae
- Mode of infection: ingestion of metacerceriae encysted on aquatic vegetation.

Life Cycle

Fasciola spp.



Clinical Manifestations

- **In acute phase:** during the migration of the larva, causes more mechanical damage. In traversing the liver tissue, it causes parenchymal injury, patients present with fever, right upper quadrant pain, eosinophilia, and tender hepatomegaly. The symptoms subside as parasites reach their final destination.
- **In chronic phase:** when flukes reside in the **bile ducts** they mature into the adult worms in about 3–4 months. patients may develop **biliary obstruction, biliary cirrhosis, obstructive jaundice, cholelithiasis, and anemia.**

- **Halzoun:** Occasionally, ingestion of raw liver of infected sheep results in a condition called halzoun (meaning suffocation). The adult worms in the liver attach to the pharyngeal mucosa, causing edematous congestion of the pharynx and surrounding areas, leading to dyspnea, acute dysphagia, deafness, and rarely, asphyxiation.

Diagnosis



- Stool Microscopy:

Demonstration of eggs in feces or aspirated bile from duodenum is the best method of diagnosis.

- Serodiagnosis:

ELISA becomes positive within 2 weeks of infection

- Imaging:

USG, CT scan, may be helpful in diagnosis.

Treatment

Oral triclabendazole is the treatment of choice.

Alternative drug is bithionol

Management and Follow-Up

Biliary Obstruction: Manual extraction of adult flukes via endoscopic retrograde cholangiopancreatography (ERCP) may be required if physical blockage occurs in the biliary tract.

Prevention

- Health education
- Preventing pollution of water courses with sheep, cattle, and human feces
- Avoid eating raw watercress or other freshwater aquatic plants grown in endemic areas
- Avoiding eating the raw liver of an infected animal (Halazone)
- Proper disinfection of watercress and other water vegetations before consumption.
- Controls intermediate host (snail)

References

- Paniker CJ (2013). Paniker's textbook of medical parasitology, New Delhi, London, Philadelphia, Panama, Jaypee Brothers Medical Publishers.
- Warren Levinson 12th Edition Review of medical Microbiology and Immunology
- Internet access for figures and illustrations.
- CDC for figures and illustrations.